

Rajalakshmi Engineering College

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2024_28_III_OOPS Using Java Lab

REC_2028_OOPS using Java_Week 4_CY

Attempt : 1
Total Mark : 40
Marks Obtained : 40

Section 1 : Coding

1. Problem Statement

A library wants to analyze book titles to count the number of words that start with an uppercase letter. This helps the library track proper nouns and important words in titles.

Your task is to write a program that, for each given title, counts and prints the number of words that start with an uppercase letter.

Input Format

The first line contains an integer T, representing the number of book titles.

Each of the next T lines contains a single title (string).

Output Format

For each title, the output print a single integer representing the number of words starting with an uppercase letter.

Refer to the sample output for formatting specifications.

Sample Test Case

Input: 1

The Chronicles of Narnia

Output: 3

Answer

```
// You are using Java
import java.util.Scanner;
class Main
{
    public static void main(String[] args)
    {
        Scanner sc=new Scanner(System.in);
        int n=sc.nextInt();
        sc.nextLine();
        for(int i=0;i<n;i++)
        {
            String s=sc.nextLine();
            int c=0;
            String[] p=s.split(" ");
            for(int j=0;j<p.length;j++)
            {
                if(Character.isUpperCase(p[j].charAt(0)))
                {
                    c++;
                }
            }
            System.out.println(c);
        }
    }
}
```

}
Status : Correct

Marks : 10/10

2. Problem Statement

In a college, students are required to create unique usernames for accessing the digital library.

The librarian needs your help to verify whether the usernames entered by students are valid.

A username is considered valid if:

It contains only letters (a–z, A–Z) and digits (0–9). Its length is between 5 and 15 characters (inclusive). It must start with a letter (not a digit).

Your task is to determine whether each username in the list is valid or not.

Input Format

The first line of input contains an integer T, representing the number of usernames to check.

The next T lines each contain a string S, representing a username.

Output Format

For each username S, the output print "YES" if it is valid.

Otherwise, the output print "NO".

Refer to the sample output for formatting specifications.

Sample Test Case

Input: 1

Alice123

Output: YES

Answer

```
// You are using Java
import java.util.Scanner;
class UsernameValidator
{
    public static void main(String[] args)
    {

        Scanner sc = new Scanner(System.in);
        int T = sc.nextInt();
        sc.nextLine();
        for (int i = 0; i < T; i++)
        {
            String s = sc.nextLine().trim();
            if (isValidUsername(s))
                System.out.println("YES");
            else
                System.out.println("NO");
        }
        sc.close();
    }
    private static boolean isValidUsername(String s)
    {
        if (s.length() < 5 || s.length() > 15)
            return false;
        if (!Character.isLetter(s.charAt(0)))
            return false;
        for (char ch : s.toCharArray())
        {
            if (!Character.isLetterOrDigit(ch))
                return false;
        }
        return true;
    }
}
```

Status : Correct**Marks : 10/10****3. Problem Statement**

A bookstore wants to analyze the titles of books to determine their longest word in each title. This helps in designing banners and covers.

Your task is to write a program that, given a sentence (book title), finds and prints the longest word. If multiple words have the same maximum length, print the first one.

Input Format

The input contains a single line containing a sentence representing the book title.

Output Format

The output prints a string representing the longest word in the sentence (book title).

Refer to the sample output for formatting specifications.

Sample Test Case

Input: The Chronicles of Narnia

Output: Chronicles

Answer

```
// You are using Java
import java.util.*;
class Main
{
    public static void main(String[] args)
    {
        Scanner sc = new Scanner(System.in);
        String sentence = sc.nextLine();
        String[] words = sentence.split(" ");
        String longest = "";
        for (String word : words)
        {
            if (word.length() > longest.length())
            {
                longest = word;
            }
        }
    }
}
```

```
}  
    System.out.println(longest);  
}  
  
}
```

Status : Correct

Marks : 10/10

4. Problem Statement

Meera is practicing her English vocabulary. She wants to focus on words that have more vowels in them, as they help improve her pronunciation. She decides to extract only those words from a sentence that contain at least two vowels.

Your task is to help Meera by writing a program that finds such words from the given sentence.

Input Format

The input contains a string representing the sentence.

Output Format

The output prints all the words that contain at least two vowels, separated by a space.

If no such word exists, print "No words with two vowels".

Refer to the sample output for formatting specifications.

Sample Test Case

Input: This is an example sentence

Output: example sentence

Answer

```
// You are using Java
```

```

import java.util.Scanner;
class TwoVowelWords
{
    public static void main(String[] args)
    {
        Scanner sc = new Scanner(System.in);
        String sentence = sc.nextLine().trim();
        String[] words = sentence.split(" ");
        StringBuilder result = new StringBuilder();

        for (String word : words)
        {
            if (countVowels(word) >= 2)
                result.append(word).append(" ");
        }
        if (result.length() > 0)
            System.out.println(result.toString().trim());
        else
            System.out.println("No words with two vowels");
        sc.close();
    }
    private static int countVowels(String word)
    {
        int count = 0;
        for (char ch : word.toLowerCase().toCharArray())
        {
            if ("aeiou".indexOf(ch) != -1)
                count++;
        }
        return count;
    }
}

```

Status : Correct

Marks : 10/10